

SIMATS ENGINEERING

CO3-ASSESSMENT TOOL 1

Experiments

COURSE NAME: Computer Networks

COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO3: *Analyze the performance of core network protocols such as TCP, UDP, HTTP, and DNS under varying conditions*

Assessment Tool Weightage: 40%

Dr.V.Parthipan

Code of Conduct:

*I **Antony Magrin J (19257103)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student Antony Magrin.J

Criteria	Understanding of Concepts 2.5	Experimental Design 3	Use of Tools 2	Analysis of Results 1.5	Documentation 1	Total (10)
Q1	2.5	3	2	1	1	9.5
Q2	2.5	3	2	1	1	9.5
Q3	2.5	3	2	1	1	9.5
Q4	2	3	2	1	1	9

Faculty Signature

Experiments

QUESTIONS:4

Total Marks:40

1. Capture packets during a TCP connection setup. Identify the three packets involved in the handshake (SYN, SYN-ACK, and ACK). Demonstrate the role of each flag and how they establish a reliable connection. (BL4) (10)
2. Capture a DNS query using UDP in Wireshark. Compare the UDP header size with TCP and note the absence of sequence numbers or acknowledgments. Discuss how this lightweight design impacts speed and reliability. (BL4) (10)
3. Simulate packet loss (e.g., disconnect briefly) and capture traffic in Wireshark. Observe how TCP retransmits lost packets while UDP does not. Explain why this difference makes TCP reliable but slower compared to UDP. (BL4) (10)
4. Use Wireshark to capture DNS traffic while resolving a domain name. Identify the query type (A, AAAA, MX) and the response received. Measure the time taken for resolution and explain why DNS typically uses UDP.(BL4) (10)

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

Wireshark interface showing packet capture details for Ethernet II, Internet Protocol Version 4, and Transmission Control Protocol. The packet list shows multiple TCP retransmissions. The packet details pane shows the structure of the captured packet, including Ethernet II, IP, and TCP headers.

No.	Time	Source	Destination	Protocol	Length	Info
1452	246.881788	172.25.50.145	172.23.29.151	TCP	66	[TCP Retransmission] [TCP Port numbers reused] 61778 → 7680 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM=1
1452	246.886369	172.25.55.218	172.23.29.232	TCP	66	[TCP Retransmission] [TCP Port numbers reused] 56113 → 7680 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM=1
1452	246.851782	172.23.17.97	172.23.164.198	TCP	66	[TCP Retransmission] [TCP Port numbers reused] 56429 → 7680 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM=1
1453	246.969749	172.19.26.23	172.23.162.115	TCP	66	[TCP Retransmission] [TCP Port numbers reused] 59898 → 7680 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM=1
1454	247.033119	172.23.163.129	172.23.20.135	TCP	66	[TCP Retransmission] [TCP Port numbers reused] 65500 → 7680 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM=1
1455	247.081287	172.25.28.11	172.23.166.103	TCP	66	61130 → 7680 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM=1
1455	247.087699	172.25.28.11	172.23.28.6	TCP	66	61135 → 7680 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM=1
1455	247.133700	172.25.53.205	172.23.28.60	TCP	66	[TCP Retransmission] [TCP Port numbers reused] 62123 → 7680 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM=1

Frame 9: 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface \Device\NPF_{33DCAC34-6EDE-4C57-BA0F-0C2424D1D35C}, id 0
Ethernet II, Src: 24:d5:e4:6d:bc:f7 (24:d5:e4:6d:bc:f7), Dst: Giga-Byt_93:54:1d (d8:5e:d3:93:54:1d)
Internet Protocol Version 4, Src: 172.25.33.201, Dst: 172.23.166.41
Transmission Control Protocol, Src Port: 62584, Dst Port: 7680, Seq: 0, Len: 0

0000 d8 5e d3 93 54 1d 24 d5 e4 6d bc f7 08 00 45 00 ...T...S...E...
0010 00 34 19 9c 40 00 7b 06 c6 04 ac 19 21 c9 ac 17 ...4...@...!
0020 a6 29 f4 78 1e 00 24 83 0e 4d 00 00 00 00 80 02 ...).x...\$...M...
0030 ff ff 09 a4 00 00 04 05 b4 01 03 03 08 01 01
0040 04 02 ..

Wireshark interface showing a packet capture on the 'ethernet' interface. The packet list pane displays several DNS queries and responses. The packet details pane shows the structure of a DNS query packet (Frame 7174).

Packet List:

No.	Time	Source	Destination	Protocol	Length	Info
3548	613.101944	172.23.164.130	172.21.100.31	DNS	88	Standard query 0xbe88 A hangouts.clients6.google.com
3548	613.102847	172.21.100.31	172.23.164.130	DNS	138	Standard query response 0x2c87 HTTPS hangouts.clients6.google.com SOA ns1.google.com
3548	613.102847	172.21.100.31	172.23.164.130	DNS	104	Standard query response 0xbe88 A hangouts.clients6.google.com A 142.251.223.234
3548	613.152813	172.21.100.31	172.23.164.130	DNS	104	Standard query response 0xa379 A meetings.clients6.google.com A 172.217.24.202
3561	615.291861	192.168.0.239	255.255.255.255	DNS	75	Standard query 0x8451 A time.google.com
3658	633.190422	172.23.164.130	172.21.100.31	DNS	80	Standard query 0xa207 HTTPS beacons.gcp.gvt2.com
3658	633.190785	172.23.164.130	172.21.100.31	DNS	80	Standard query 0x7d48 A beacons.gcp.gvt2.com
3658	633.191523	172.21.100.31	172.23.164.130	DNS	167	Standard query response 0xa207 HTTPS beacons.gcp.gvt2.com CNAME beacons-handoff.gcp.gvt2.com SOA ns1.google.com
3658	633.191715	172.21.100.31	172.23.164.130	DNS	126	Standard query response 0x7d48 A beacons.gcp.gvt2.com CNAME beacons-handoff.gcp.gvt2.com A 74.125.136.94

Packet Details (Frame 7174):

- Frame 7174: 80 bytes on wire (640 bits), 80 bytes captured (640 bits) on interface \Device\NPF_{33DCAC34-6EDE-4C57-BA0F-0C2424D1D35C}, id 0
- Ethernet II, Src: 74:56:3c:21:fe:bd (74:56:3c:21:fe:bd), Dst: 24:d5:e4:6d:bc:f7 (24:d5:e4:6d:bc:f7)
- Internet Protocol Version 4, Src: 172.23.164.130, Dst: 172.21.100.31
- User Datagram Protocol, Src Port: 56857, Dst Port: 53
- Domain Name System (query)

Packet Bytes:

```
0000  24 d5 e4 6d bc f7 74 56 3c 21 fe bd 00 00 45 00  $ m EV < ! ... E
0010  00 42 fb c8 00 00 00 11 00 00 ac 17 a4 81 ac 15  B ...
0020  64 1f de 19 00 35 00 2e 61 0e 67 c8 01 00 00 01  d ... S , a g
0030  00 00 00 00 00 00 07 62 65 61 63 6f 6e 73 03 67  ..... b eacons g
0040  63 70 04 67 76 74 32 03 63 6f 6d 00 00 41 00 01  cp gvt2 com A
```

Wireshark interface showing a packet capture on the 'tap' interface. The packet list displays several TCP retransmissions and SYN packets. The selected packet (No. 4) is a TCP segment with details expanded to show Ethernet II, Internet Protocol Version 4, and Transmission Control Protocol fields.

No.	Time	Source	Destination	Protocol	Length	Info
70589	132.031277	172.23.162.178	172.23.164.129	TCP	66	[TCP Retransmission] [TCP Port numbers reused] 53944 → 7680 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM=1
70594	132.034453	172.23.55.220	172.23.164.172	TCP	66	51247 → 7680 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM=1
70599	132.061126	172.23.55.220	172.23.162.141	TCP	66	51249 → 7680 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM=1
70600	132.061126	172.23.55.220	172.23.162.17	TCP	66	51253 → 7680 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM=1
70606	132.071350	172.23.27.127	172.23.164.122	TCP	66	64701 → 7680 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM=1
70611	132.083067	172.23.26.102	172.23.28.6	TCP	66	[TCP Retransmission] [TCP Port numbers reused] 49817 → 7680 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM=1
70629	132.154413	172.23.164.134	172.23.30.252	TCP	66	61165 → 7680 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM=1
70644	132.187181	172.23.26.242	172.23.163.96	TCP	66	60521 → 7680 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM=1

Frame 4: 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface \Device\NPF_{33DCAC34-6EDE-4C57-BA0F-0C2424D1D35C}, id 0
> Ethernet II, Src: 24:d5:e4:6d:bc:f7 (24:d5:e4:6d:bc:f7), Dst: 74:56:3c:06:99:21 (74:56:3c:06:99:21)
> Internet Protocol Version 4, Src: 172.23.163.220, Dst: 172.23.26.27
> Transmission Control Protocol, Src Port: 58347, Dst Port: 7680, Seq: 0, Len: 0

0000 74 56 3c 06 99 21 24 d5 e4 6d bc f7 08 00 45 00 tv<...!\$>...m...E<
0010 00 34 71 d4 40 00 7f 06 73 c9 ac 17 a3 dc ac 17 4q @...s.....
0020 1a 1b e3 eb 1e 00 16 86 ad 35 00 00 00 00 00 02-S.....
0030 ff ff 93 42 00 00 02 04 05 b4 01 03 03 08 01 01 ...B.....
0040 04 02 ..

Wireshark interface showing a packet capture on the Ethernet interface. The filter is set to `dns.qry.type==1||dns.qry.type==28||dns.qry.type==15`.

No.	Time	Source	Destination	Protocol	Length	Info
1000	196.366908	172.23.17.221	224.0.0.251	MDNS	70	Standard query 0x0000 AAAA wpad.local, "QU" question
1000	196.370521	fe80::26da:f0cd:5e8... ff02::fb		MDNS	90	Standard query 0x0000 AAAA wpad.local, "QU" question
1000	196.376219	fe80::26da:f0cd:5e8... ff02::fb		MDNS	90	Standard query 0x0000 A wpad.local, "QM" question
1000	196.449712	169.254.125.153	224.0.0.252	LLMNR	66	Standard query 0xc43b A isatap
1001	196.723788	172.23.30.184	224.0.0.251	MDNS	102	Standard query 0x0000 AAAA SAIL303-PC05.local, "QM" question A SAIL303-PC05.local, "QM" question
1005	197.939223	172.23.30.184	224.0.0.251	MDNS	102	Standard query 0x0000 A SAIL504-PC14.local, "QM" question AAAA SAIL504-PC14.local, "QM" question
1009	198.750379	172.23.30.184	224.0.0.251	MDNS	102	Standard query 0x0000 A SAIL201-PC16.local, "QM" question AAAA SAIL201-PC16.local, "QM" question
1014	199.917190	172.23.30.184	224.0.0.251	MDNS	102	Standard query 0x0000 A SAIL409-PC21.local, "QM" question AAAA SAIL409-PC21.local, "QM" question

Frame 47: 104 bytes on wire (832 bits), 104 bytes captured (832 bits) on interface \Device\NPF_{33DCAC34-6EDE-4C57-BA0F-0C2424D1D35C}, id 0

Ethernet II, Src: IntelCor_59:11:22 (3c:21:9c:59:11:22), Dst: IPv4mcast_fb (01:00:5e:00:00:fb)

Internet Protocol Version 4, Src: 172.23.30.184, Dst: 224.0.0.251

User Datagram Protocol, Src Port: 5353, Dst Port: 5353

Multicast Domain Name System (query)

Hex dump of the packet data:

```
0000 01 00 5e 00 00 fb 3c 21 9c 59 11 22 08 00 45 00 .....!..Y...E..
0010 00 5a 68 4f 00 00 01 11 a5 79 ac 17 1e b8 e0 00 ..ZhO....y.....
0020 00 fb 14 e9 14 e9 00 46 d8 8c 00 00 00 00 00 02 .....F.....
0030 00 00 00 00 00 00 0d 53 41 49 4c 31 30 31 2d 50 .....S AIL101-P
0040 43 31 33 34 05 6c 6f 63 61 6c 00 00 1c 00 ff 0d C134-loc al.....
0050 53 41 49 4c 31 30 31 2d 50 43 31 33 34 05 6c 6f SAIL101- PC134-lo
0060 63 61 6c 00 00 01 00 ff cal.....
```